

# DATA INTEGRY REPORT

**Data set:**task1\_30806020201621\_combined\_data.csv

**The combined data that i had made contains of :**

- 435 records
- 13 columns
- alot of problems

## **MISSING\_ VALUES:**

**What i found:**

1. A lot of missing values in serval columns
2. Some members has no checkout recorded or a book had not been returned.

**Where is the missing values:**

In serval columns (: first\_name, last\_name, grade, neighborhood, membership\_status, join\_date, checkout\_id, book\_id, checkout\_date, return\_date, title, and author.)

**How i cloud solve it by explain:**

1. I treated missing values of string columns like(first\_name,last\_name,neighborhhod,membership\_status,join\_date,checkout\_date,return\_date,title and author) by using mode () method .
  2. I treated missing values of integer columns like: **grades** by using median bec it doesn't pulled by outliers also thetis data set has no outlier.
  3. I tread columns:checkout\_id ,book\_id by filling their missing values by words;(No checkout, No books).
-

## ***DUPLICATES RECORD:***

***What i found:*** i found 8 exact duplicates rows in the data. They were not new or different events.

**What I did:** I deleted these 8 duplicate row

**Why:** To keep only unique and correct records, without deleting any real data.

---

## **Inconsistent Text Values :**

***Where i found:***

- In neighborhood columns as : ('Maadi ', 'zamalek', 'HELIOPOLIS', 'Nasr City', and 'NASR CITY')
- In membership statues as : ('Active', 'active', 'Inactive', 'inactive', and 'INACTIVE'.)

***Their records:*** in member ship their was 79 but in neighbourhood their was 31

***How i could fix it:***

1. If fix member ship statues by function `str.string()` ,`str.strip()`
  2. I fix neighbourhood by function `str.String()` , `str.strip()`
  3. Why i uses this functions to remove extra spaces before the word and to make the first letter is capital and the letters follow them is small to become all in one formats
- 

## **Invalid Member IDs :**

- ***What I found:*** I found 5 records with wrong Member IDs (like 1104, 1150, 1201). These IDs are not in our official member list (1001–1080).
- ***What I did:*** I deleted these 5 wrong records.
- ***Why:*** Because we don't know who these records belong to. Deleting them is safer than guessing the correct numbers.

